

Pelvic Outlet Diameter Medial Axis Calculation

2D Point Annotator

Current application implementation

Purpose

This note describes how the application now calculates the medial axis used to construct the pelvic outlet diameter (POD) line. The calculation is performed in the 512 by 512 manual contour mask space, independent of the displayed image size or original image resolution. The resulting construction points are converted back to image coordinates for viewing and for the derived POD line annotation.

Inputs and Output

Input	Required	Role
L-C-PO	yes	left pelvic outlet contour, drawn on image right
R-C-PO	yes	right pelvic outlet contour, drawn on image left
L-LPO	yes	derived left lateral pelvic outlet point
R-LPO	yes	derived right lateral pelvic outlet point
L-C-PS	optional pair	left pubic symphysis contour evidence for the axis
R-C-PS	optional pair	right pubic symphysis contour evidence for the axis

The output is the derived POD line segment. Its endpoints are selected on the left and right pelvic outlet contours, but the centerline used to orient that segment is fit from both the pelvic outlet contours and, when available, the pubic symphysis contours.

Coordinate Convention

Protocol side labels are patient-side labels. On the displayed AP image, the patient's left side appears on image right, and the patient's right side appears on image left. Therefore, for this POD calculation:

- the medial edge of L-C-PO or L-C-PS is the smallest active x value on a row;
- the medial edge of R-C-PO or R-C-PS is the largest active x value on a row.

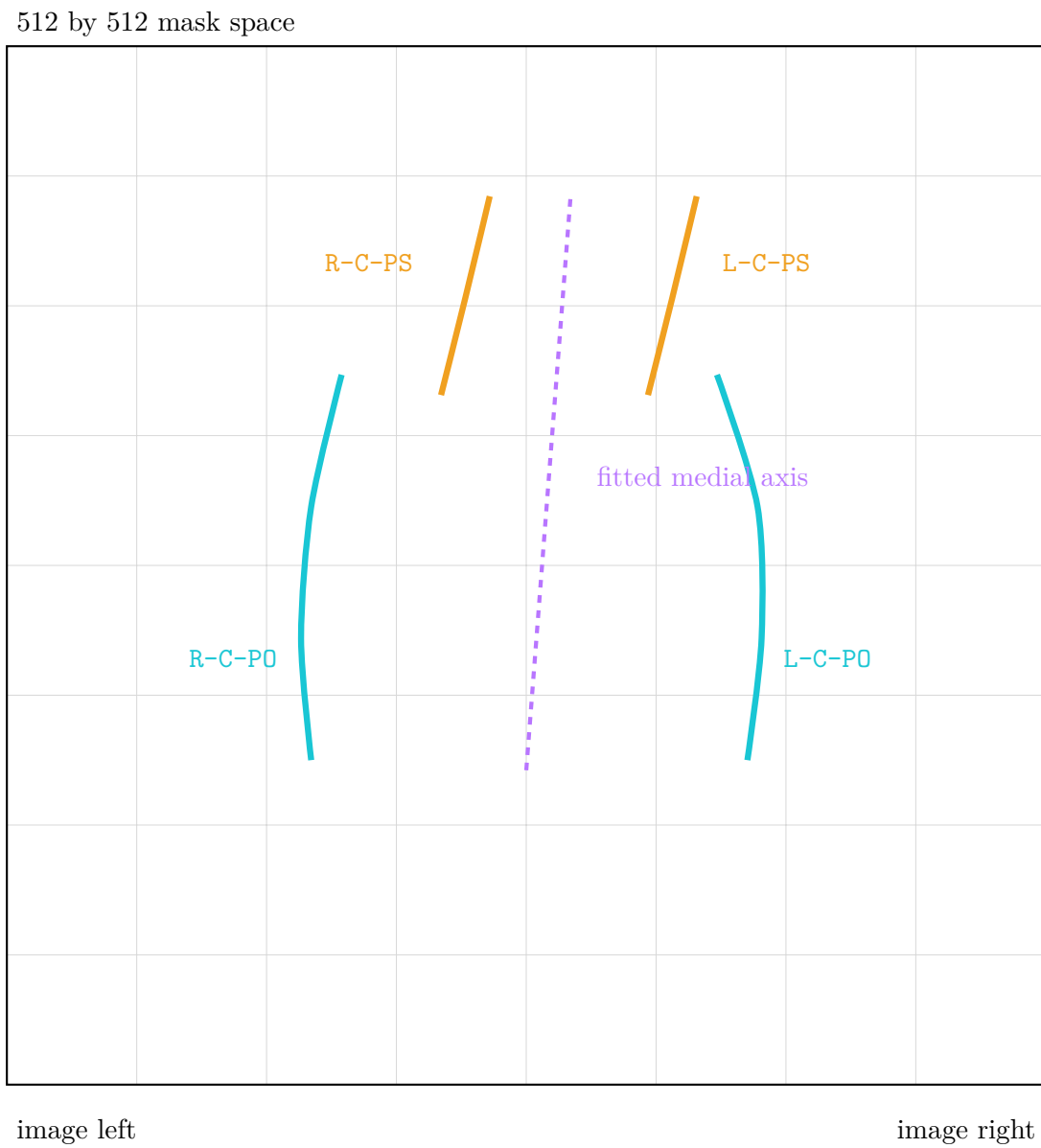


Figure 1: The axis is fit in fixed 512 by 512 mask coordinates. The PO contours are required. The PS contours contribute extra lower-midline evidence when both sides are present.

Step 1: Collect Row-Wise Midpoint Samples

For each contour pair, the application scans mask rows. On a row y , let L_y be the set of active x pixels in the left-side contour mask and R_y be the set of active x pixels in the right-side contour mask.

$$x_L^m(y) = \min L_y, \quad x_R^m(y) = \max R_y$$

The row is accepted only when both contours have active pixels on that row and the left medial edge is to the right of the right medial edge:

$$x_L^m(y) > x_R^m(y)$$

The medial-axis sample for the row is the midpoint between those two medial edges:

$$m(y) = \left(\frac{x_L^m(y) + x_R^m(y)}{2}, y \right)$$

This is done for L-C-P0/R-C-P0. If both pubic symphysis masks exist, the same sampling rule is also applied to L-C-PS/R-C-PS. The final sample set is:

$$S = S_{PO} \cup S_{PS}$$

where S_{PS} is included only when it can be generated from both PS masks. If exact shared rows do not provide at least two samples for a pair, the implementation falls back to nearest-row pairing within a tolerance of $\max(8, \text{round}(0.05H))$, where H is the mask height.

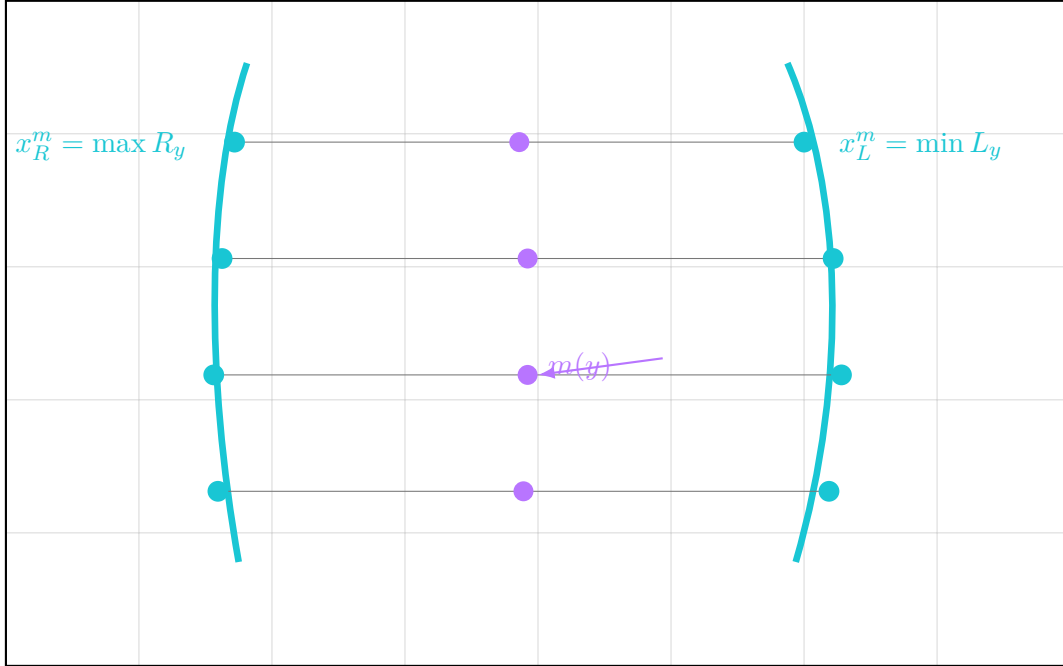


Figure 2: Each accepted row contributes the midpoint between the two medial contour edges. The same rule is used for the pelvic outlet pair and the pubic symphysis pair.

Step 2: Fit the Medial Axis

The sample points S are fit with a straight line using principal component analysis. In implementation, this is done with singular value decomposition.

$$o = \frac{1}{|S|} \sum_{s_i \in S} s_i$$

The centered sample matrix C contains rows $s_i - o$. The first right singular vector of C is the unit axis direction d . The direction is flipped if needed so that $d_y \geq 0$, giving a stable top-to-bottom axis.

For each sample:

$$t_i = (s_i - o) \cdot d$$

The visualized axis segment spans from:

$$o + d \min_i(t_i) \quad \text{to} \quad o + d \max_i(t_i)$$

Step 3: Place the POD Cross-Line

The lateral pelvic outlet points are projected onto the medial axis:

$$t_L = (p_L - o) \cdot d, \quad t_R = (p_R - o) \cdot d$$

where p_L is L-LP0 and p_R is R-LP0, both represented in the 512 by 512 mask coordinate system. The POD center is halfway between those two projected locations, clamped to the fitted sample extent:

$$t_{mid} = \text{clamp} \left(\frac{t_L + t_R}{2}, \min_i(t_i), \max_i(t_i) \right)$$

$$a_{mid} = o + dt_{mid}$$

The POD direction is perpendicular to the fitted medial axis:

$$n = (-d_y, d_x)$$

Starting from a_{mid} , the application searches each pelvic outlet contour for an endpoint on the appropriate side of this normal. It searches in expanding bands around the perpendicular cross-section and picks the outward-most contour pixel for each side. The returned POD annotation is the line segment between those two selected pelvic outlet contour pixels.

Effect of the Pubic Symphysis Contours

Before the latest change, the medial axis was fit only from L-C-P0/R-C-P0. Now, when L-C-PS and R-C-PS are both annotated, their row-wise midpoint samples are added to the same fit. This lets the lower pelvic midline influence the axis, which is useful when the two outlet contours alone produce a centerline that is visually offset from the overall bilateral symmetry.

The PS contours do not replace the outlet contours. The endpoint search for the POD line still uses L-C-P0 and R-C-P0; the PS contours only improve the fitted medial axis used to choose the POD cross-section and angle.

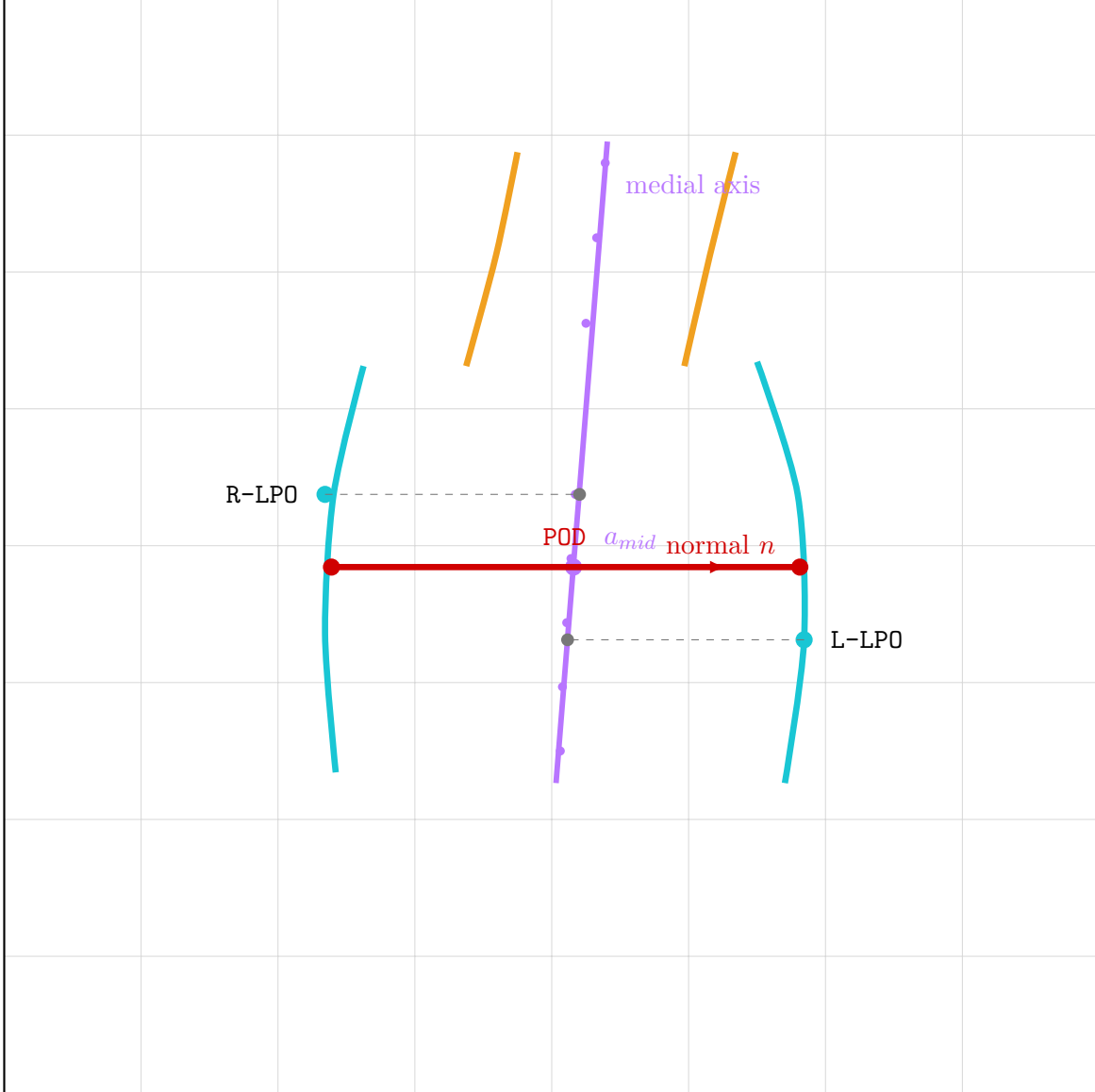


Figure 3: The LPO points are projected onto the medial axis. The halfway point between those projections defines the cross-section. The final POD line is the perpendicular segment from the right pelvic outlet contour to the left pelvic outlet contour.

Failure Cases

The application does not generate a POD line when:

- either pelvic outlet contour mask is missing or empty;
- either lateral pelvic outlet point is unavailable;
- the PO and optional PS contour pairs do not produce at least two valid midpoint samples in total;
- the SVD line fit is degenerate;
- a POD endpoint cannot be selected from either pelvic outlet contour.

Implementation Location

The current implementation is in `app/src/main.py`:

- `_protocol_pod_medial_axis_samples`: builds row-wise midpoint samples for one left/right contour pair;
- `_fit_protocol_pod_medial_axis`: fits the straight medial axis using SVD;
- `_protocol_pod_perpendicular_endpoint`: chooses the PO contour endpoint for one side of the POD line;
- `_derive_protocol_pod`: combines PO and optional PS samples and returns the final POD segment.